

JOURNAL PUBLICATIONS

S.No.	Title	Journal Name/ (National/ International)	Impact factor/ Indexing	Status/ Vol, Year, Pag.No.	Document link
1	Advances in Electrochemical CO ₂ Reduction to CO for Enabling A Sustainable Energy Economy	Progress in Materials Science/ International	40.8/SCI, Scopus & Web of Science	Published/ May 2026: 101655	View article
2	Towards Carbon Neutrality: Challenges and Advances in the Electrochemical Selective Reduction of Carbon Dioxide into Methane	Progress in Energy and Combustion Science	37.0/SCI, Scopus & Web of Science	Reviews completed	
3	Advancements and Perspective of Environmentally Sustainable Technologies for Electrochemical Selective Conversion of CO ₂ to Methanol	Catalysis Reviews/ International	13.6/ SCI, Scopus & Web of Science	Published/ May 2024: 1-113	View article
4	Unraveling the in-situ engineered mixed metal oxides anchored on activated carbon cloth as a flexible electrode material for electrochemical reduction of CO ₂ into formic acid	Sustainable Materials and Technologies/ International	8.7/SCI, Scopus & Web of Science	Published/ Vol 43, April 2025: e01260	View article
5	Unlocking the potential of in-situ thermally reduced graphene oxide anchored with ternary metal oxide-spinel catalyst for sustainable high-performance electrochemical valorisation of CO ₂	Fuel (Elsevier)/ International	6.7/ SCI, Scopus & Web of Science	Published/ Vol 387, May 2025: 134458	View article
6	Unleashing the potential of N-doped metal-free bi-functional hollow carbon spheres: Catalyst for environmental conservation and advancing sustainable energy solutions	Fuel (Elsevier)/ International	6.7/ SCI, Scopus & Web of Science	Published/ Vol 374, Oct 2024: 132457	View article

7	Recent Advances in the Fixation of CO ₂ into Quinazoline and Benzimidazole	Energy and Fuels (ACS)/ International	5.3/ SCI, Scopus & Web of Science	Published/ Vol 38 (10), April 2024: 8481-8515	View article
8	Structure-property connections in dual-phase multifunctional barium titanate based nanocomposites: Insights from charge density distributions	Materials Chemistry and Physics/ International	4.6/ SCI, Scopus & Web of Science	Published/ Vol 318, May 2024: 129290	View article
9	Multifunctionality of AlBaFe ₁₂ O ₁₉ /CoZnFe ₂ O ₄ hybrid nanocomposite: Promising structural, elastic, morphological, compositional, optical, and magnetic properties	Journal of Physics and Chemistry of Solids/ International	4.3/ SCI, Scopus & Web of Science	Published/ Vol 174, Mar 2023: 111134	View article
10	In-Situ MgO engineered dendritic fibrous nano silica as an efficient catalyst for fixation of CO ₂ at atmospheric pressure	Chemical Engineering Science/ International	4.3/ SCI, Scopus & Web of Science	Published/ Vol 317, Nov 2025:122068	View article
11	Optical Bandgap characterization of Novel Nano scale CBM-CZF composite	Bulletin of Environment, Pharmacology and Life Sciences (BEPLS)/ International	4.0/ Web of Science & UGC	Published/ Special Issue [1], May 2022: 643-650	View article
12	Deep Eutectic Solvent Engineered Dendritic Fibrous Nano-Silica Catalyst for Sustainable Fixation of CO ₂ Into Value-Added Product at Atmospheric Condition	Applied Organometallic Chemistry/ International	3.7/ SCI, Scopus & Web of Science	Published/ Vol 39, Jan 2025:e7910	View article
13	Investigating structure, elasticity, morphology, composition, and optical behavior of Al-doped BaFe ₁₂ O ₁₉ /CoZnFe ₂ O ₄ hybrid composite	Journal of Materials Research/ International	2.9/ SCI, Scopus & Web of Science	Published/ Vol 38, Jan 2023:1239 –1253	View article

14	Electron density effect of hybrid ferrites on structural, spectroscopic, optical, and magnetic properties in $(\text{BaTiO}_3)_{(1-x)} + (\text{ZnFe}_2\text{O}_4)_x$ nanoceramics	Journal of Material Science: Materials in Electronics/ International	2.8/ SCI, Scopus, & Web of Science	Published/ / Vol 34, Jan 2023: 104	View article
15	Potential biological and optoelectronic applications of AgO:ZnO nanocomposite synthesized by green approach	The European Physical Journal Plus/ International	2.8/SCI, Scopus, & Web of Science	Published/ / Vol 139, Dec 2024: 118	View article
16	Investigating Optical Properties of Novel Heterostructure Aluminium Doped Barium Hexaferrite (AlBaM) Cobalt Zinc Ferrite (CZFO) Nanocomposite.	Indian Journal of Natural Science/ National	2.74/ Web of Science & UGC	Published/ / Vol 13 Issue 71, Apr 2022: 39368-39375	View article
17	A Topological Control Algorithm for Decision-Making in Complex Material Systems	Indian Journal of Natural Science/ National	2.74/ Web of Science & UGC	Published/ / Vol 15 Issue 85, Aug 2024: 1-11	View article
18	Synthesize of ZnO and CuO nanoparticles with plasma jet at different treatment times and testing its optical parameters with UV-Vis-NIR	Applied Physics A: Materials Science & Processing (Springer)/ International	2.6/ SCI, Scopus, & Web of Science	Published/ / Vol 130, Jul 2024: 533	View article
19	Biological activity of MgO nanoparticle synthesis by plasma-assisted reduction method	Physica Scripta (IOP Science)/ International	2.6/ SCI, Scopus, & Web of Science	Published/ / Vol 99, Number 11 Oct 2024: 115901	View article
20	An Efficient and Selective Synthesis of Chalcones via Claisen-Schmidt Condensation Reaction Catalyzed by CuO@rGO Nano Catalyst	Catalysis Letters/ International	2.4/ SCI, Scopus, & Web of Science	Published/Vol 155, Sep 2025: 327	View article
21	Comprehensive Analysis of Optical, Dielectric and Electrical properties in Al- and Cu- Doped Barium	Journal of Electronic Materials/ International	2.1/ SCI, Scopus, & Web of Science	Published/ / Vol 53 Issue 06, Jun 2024:	View article

	Hexaferrite/Cobalt-Zinc Ferrite Hybrid Nanocomposites			39368-39375	
22	Deliberating Structural, Optical and Spectroscopic Properties of novel Barium Titanate Zinc Ferrite nanocomposite synthesized using low temperature techniques	Stochastic modelling and applications/ National	1.35/ UGC Care Group I & Publon	Published/ / Vol 26 Issue 03, Jan 2022: 598-605	View article
23	Studying Structural and Optical Properties of Novel Aluminium Barium Hexaferrite (AlBaM) Cobalt Zinc Ferrite (CZFO) Nanocomposite.	Stochastic modelling and applications/ National	1.35/ UGC Care Group I & Publon	Published/ / Vol 26 Issue 03, Jan 2022: 606-612	View article
24	Qualitative Analysis of Magnetic Moment in BaTiO ₃ -NiFe _{3-x} O ₄ Nanocomposite Synthesized by Novel low Temperature technique	International Journal of Recent Technology and Engineering (IJRTE)/ International	1.2/ Publon (Web of Science) & UGC	Published/ Vol 10 Issue 05, Jan 2022 : 17-22	View article
25	Study of Bandgap Energy of Novel Cu _{0.5} Ba _{0.5} Fe ₁₂ O ₁₉) _{1-x} /Co _{0.6} Zn _{0.4} Fe ₂ O ₄) _x Nanocomposite	International Journal of Recent Technology and Engineering (IJRTE)/ International	1.2/ Publon (Web of Science) & UGC	Published/ Vol 10 Issue 04, Nov 2021 : 171-176	View article
26	Novel Heterostructure BTO-ZFO nanocomposite: A Quantitative Study	Rasayan Journal of Chemistry/ National	0.33/ Scopus & Web of Science	Published/ / Vol 15 Issue 04, Dec 2022: 2197-2202	View article
27	Ferrites: A study on structure, properties, synthesis, characterization, and Applications	Journal of Xidian University/ International	5.4/ Scopus & UGC Care	Published/ / Vol 17 Issue 07, Aug 2023: 1474 – 1526	View article

Research Patents

S.No.	Patent Title	Name of the Applicant(s)	Filed/Granted Patent No.	Award Date
1	Insitu Engineering mixed metal oxides on functionalized carbon cloth: A Paradigm for Flexible Electrode Material in CO ₂ Electroreduction into Formic Acid.	<u>Dr. Allwin Sudhakaran</u> , Dr. Arvind H. Jadhav, Hemavathi M, Dr. Akshaya K. Samal, Mr. Rohan Mugadum	JBR.1583	Applied on 12-08-2024

PUBLICATIONS IN CONFERENCE

S.No.	Title	Journal Name/ (National/ International)	Impact factor/ Indexing	Status/ Vol, Year, Pag.No.	Document link
1	Solar energy-based LED-powered light and inverter circuit with different types of sensors.	IOP Conference Series: Material Science and Engineering/ International	0.5/ Scopus	Published/ / Vol 1126, Mar 2021	View Link
2	Ferrites a study on structure, Properties, Synthesis and Characterization and Application	ISBN book Chapter in Conference Proceedings / International	—	Published/ / OP-206, Jul 2021: 301	View Link
3	A Review of Factors affecting Structural, Morphological, and Dielectric Properties of Barium Titanate Using Different Preparation Techniques	ISBN book Chapter in Conference Proceedings / International	—	Published/ / OP-209, Jul 2021: 304	View Link

BOOK PUBLICATIONS

S. No.	Title	Area/ Year	Publisher	ISBN No.
1	Basics of Major Synthesis and Characterization of Nanomaterials	Nanomaterials/ 2021	KDP Publishers	ISBN-13 : 979-8723281752
2	Fundamentals of Zeolites & Composites : Nanoscience and Nanotechnology	Nanoscience and Nanotechnology/ 2021	KDP Publishers	ISBN-13 : 979-8746228833
3	Chemistry behind Nanoscience & Nanotechnology	Nanochemistry/ 2021	KDP Publishers	ISBN-13: 979-8806630927
4	Basic Fundamentals of Graphene & Graphene sheets based Nanocomposites	Nanotechnology/ 2021	KDP Publishers	ASIN: B094T2NTV5
5	A Review on Mesoporous materials & their Nanocomposites	Nanoscience and Nanocomposites/ 2021	KDP Publishers	ASIN: B094SQSX1C
6	Briefing various Modeling techniques of polymer/ CNT Nanocomposites	Nanocomposite/ 2021	KDP Publishers	ASIN: B094R1RPDZ
7	Introduction to Cellulose, Hemicellulose & Lignin based Nanocomposites	Nanoscience and Nanocomposites/ 2021	KDP Publishers	ISBN-13: 979-8500770981
8	An Overview of CNT & their Nanocomposites	Nanoscience and Nanotechnology/ 2021	KDP Publishers	ASIN: B094S92NY5
9	Briefing Metal-Organic Frameworks (MOF) & its Composites	Material Science/ 2021	KDP Publishers	ASIN: B094T6B2G2

